



RESEARCH

& INNOVATION UPDATES

Drones and lasers team up to detect greenhouse gas leaks

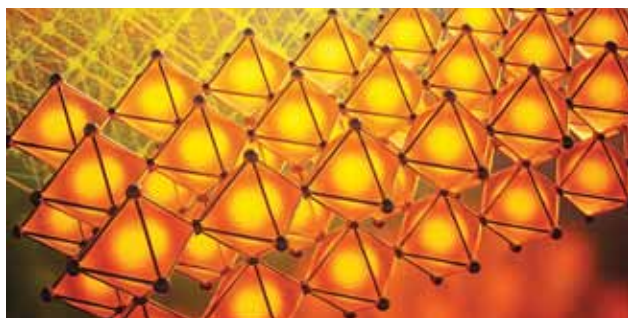


The retroreflector mounted onto the drone (Credit: Bumper DeJesus, Princeton University)

Researchers at Princeton University have developed a laser based sensing method that can locate and quantify greenhouse gas leaks, both big and small, localising emission sources to within a metre for fast repairs. The team's method uses a retroreflector-equipped drone and gas sensors at a base station to locate and measure gas leaks by bouncing laser beams. Mounting gas sensors on drones has limitations due to weight and safety concerns in hazardous environments. The researchers used a mobile base station for gas sensing and a smaller drone with a mirror. This approach uses cheaper, longer-lasting drones to monitor entire natural gas facilities in one flight, bypassing drone limitations. The method could enable simultaneous measurement of multiple gasses—a challenge in other drone based approaches due to size and power restrictions.

Machine learning helps discover high-efficiency solar cell materials

Researchers at the University of California, Davis College of Engineering, have deployed machine learning to discover high-efficiency solar cell materials by accurately predicting dynamic behaviour via fewer experiments. Perovskite devices degrade faster than silicon in harsh environments. Finding efficient and durable options is crucial due to their ABX₃ structure with many possible combinations, making trial-and-error techniques difficult. Evaluating them in different environmental conditions leads to a vast hyperparameter space. The researchers use machine learning algorithms to predict perovskite material degradation from moisture. A high-throughput system measures the photoluminescence of five perovskite films under summer conditions in Sacramento, generating a reliable training set with 7,000+ weekly measurements. Three algorithms were



(Credit: ACS Energy Letters, 2023)

trained: linear regression, neural network, and SARIMAX. SARIMAX had the best performance, matching lab results 90% over 50+ hours.